



INFERENCE SYSTEMS

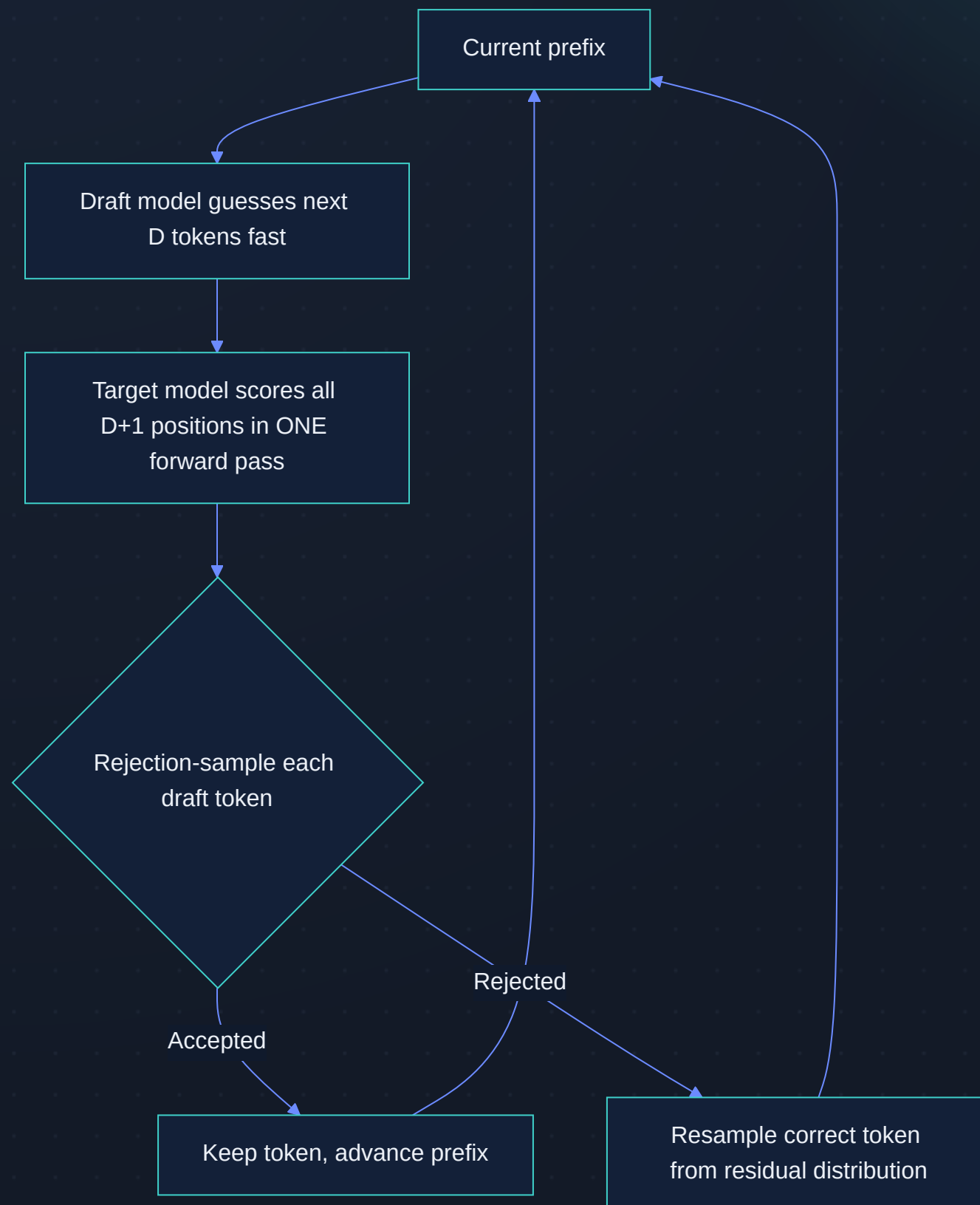
Autoregressive decoding is memory-bound, not compute-bound.

Speculative decoding fixes the math — and provably never changes a single token the model would have said.





Draft, Then Verify — In One Pass



Why Quality Never Drops: Modified Rejection Sampling

```
for i, (tok, q_i) in enumerate(zip(draft_toks,
draft_probs)):
    p_i = target_probs[i][tok]
    if random.random() < min(1.0, p_i / q_i):
        accept(tok)          # keep it, advance
        continue

    # Reject: resample from the residual, not the
    # target's argmax, so the draw stays ~ p_target.
    residual = clip(target_probs[i] - draft_probs[i], 0)
    residual /= residual.sum()
    return sample(residual), "stop"

# All accepted: target scored one extra position
# for free in the same pass – take it as a bonus.
return sample(target_probs[-1]), "continue"
```



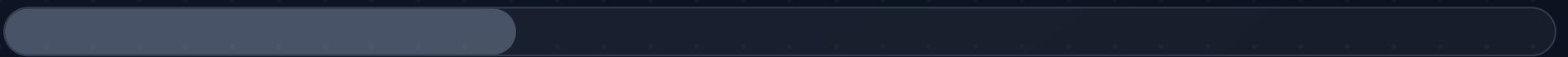


~3x

Faster, zero quality loss

Google Research's original result: a 60M-parameter T5-small draft model verified against an 11B-parameter T5-XXL target hit ~3x faster inference on translation and summarization — with outputs mathematically guaranteed identical to running the 11B model alone.

Plain autoregressive decoding – 1x baseline



Speculative decoding – ~3x throughput



Where This Breaks Without Care



Draft and target must share a tokenizer — rejection sampling operates token-by-token with no fallback for mismatched vocabularies.



A weak draft model tanks the acceptance rate: you pay for two forward passes and gain nothing over plain decoding.



A separate draft model means a second KV cache to manage — EAGLE-style self-drafting avoids this by reusing the target model's own hidden states instead of running an independent model.





Follow Sandesh Kale

for more deterministic AI systems breakdowns — the math behind why your inference stack is fast or slow.



Sandesh Kale
GenAI Solutions Architect